

Geometry-Survival Evidence Gates for Reproducible and Cost-Aware Quantum Kernel Applications Under Noise

Marcelo Claro Laranjeira

Projeto Fundamentos Quânticos Aplicados à Pesquisa em IA

Protocol frozen: 16 August 2026

SHA-256: 89c08f07f2f03a56c144bd4c4b7eddb71813bb1b043bb90920de8e5e70140eaf

Abstract

The deployment of quantum kernel methods in machine learning faces a persistent gap between theoretical promise and empirical validation under realistic noise conditions. While quantum kernels computed via feature maps on quantum hardware may capture geometric structures inaccessible to classical kernels, the conditions under which this potential translates into predictive superiority remain poorly specified. This study introduces the **Geometry-Survival Evidence Gate**, a prospective, sequential decision framework that integrates paired corrected effect estimation, equivalence testing, geometric survival analysis across fidelity tiers (ideal statevector, shot-based estimation, and depolarised noise), and computational cost accounting before authorising progression to full quantum processing unit (QPU) classification. The framework was pre-specified via SHA-256 hash and evaluated on four tabular datasets (synthetic moons, Iris binary, Wine binary, and Breast Cancer Wisconsin) using nested repeated cross-validation (4 outer folds $\times$ 3 repetitions) with Nadeau–Bengio correction for dependent folds. The primary outcome—paired difference in balanced accuracy between a fidelity quantum kernel SVM and a tuned SVM-RBF—yielded a mean ΔBAC of -0.217 (95% corrected CI $[-0.285; -0.148]$; one-sided corrected $p = 0.9999$), indicating statistical inferiority of the quantum kernel under this protocol. Equivalence testing (TOST, margin ± 0.02) yielded $p = 0.9999$, and the permutation test of signs yielded $p = 0.0005$. The effective kernel rank was 27.9% of capacity, and kernel–target alignment was 0.110. The validity ladder demonstrated geometric survival across all three fidelity tiers (BAC stable at 0.625), yet this survival did not translate into predictive advantage. The multi-dataset exploratory suite confirmed consistent RBF superiority across all 16 folds (mean $\Delta\text{BAC} = -0.146$). The Evidence Gate classified this protocol as **inconclusive for advancement to QPU**, restricting future hypotheses and providing a transparent, reproducible negative result for quantum kernel evaluation on tabular data.

Keywords: quantum kernel methods; machine learning; nested cross-validation; evidence gate; negative result; reproducibility; noise characterization

Resumo

O emprego de métodos de kernel quântico em aprendizado de máquina enfrenta uma lacuna persistente entre a promessa teórica e a validação empírica sob condições realistas de ruído. Embora kernels quânticos computados via feature maps em hardware quântico possam capturar geometrias inacessíveis a kernels clássicos, as condições sob as quais esse potencial se traduz em superioridade preditiva permanecem mal especificadas. Este estudo introduz o **Geometry-Survival Evidence Gate**—framework prospectivo e sequencial de decisão que integra estimativa de efeito pareado corrigido, teste de equivalência, análise de sobrevivência geométrica em camadas de fidelidade (statevector ideal, estimativa por shots e ruído despolarizante) e contabilidade de custo computacional antes de autorizar o avanço para classificação completa em processador quântico (QPU). O protocolo foi pré-registrado via hash SHA-256 e avaliado em quatro bases de dados tabulares (moons sintético, Iris binário, Wine binário e Breast Cancer Wisconsin) utilizando validação cruzada aninhada repetida (4

folds externos $\times$ 3 repetições) com correção de Nadeau–Bengio para folds dependentes. O desfecho primário—diferença pareada de acurácia balanceada entre SVM com kernel quântico de fidelidade e SVM-RBF sintonizado—produziu Δ BAC médio de $-0,217$ (IC95% corrigido $[-0,285; -0,148]$; p unicaudal corrigido $= 0,9999$), indicando inferioridade estatística do kernel quântico neste protocolo. O teste de equivalência (TOST, margem $\pm 0,02$) retornou $p = 0,9999$ e o teste de permutação de sinais retornou $p = 0,0005$. O posto efetivo do kernel foi de 27,9% da capacidade e o alinhamento kernel-alvo foi de 0,110. A escada de validade demonstrou sobrevivência geométrica nas três camadas de fidelidade (BAC estável em 0,625), porém essa sobrevivência não se converteu em vantagem preditiva. A suíte exploratória multibase confirmou superioridade consistente do RBF em todos os 16 folds (Δ BAC médio $= -0,146$). O Evidence Gate classificou este protocolo como **inconclusivo para avanço a QPU**, restringindo hipóteses para dados tabulares com feature maps de baixa capacidade e demonstrando o Evidence Gate como mecanismo de controle de qualidade para pesquisa em QML.

Palavras-chave: métodos de kernel quântico; aprendizado de máquina; validação cruzada aninhada; gate de evidência; resultado negativo; reprodutibilidade; caracterização de ruído

1 Introduction

Quantum machine learning (QML) has been presented as one of the most promising applications of near-term quantum computing, with claims that quantum kernels could capture geometric structures in high-dimensional spaces inaccessible to classical approaches (Bowles et al., 2024; Heyraud et al., 2022). The central premise is that a quantum feature map—a parametrised sequence of quantum gates that encodes classical data into quantum states—can produce similarity matrices (kernels) that, when fed into a classical classifier such as the Support Vector Machine (SVM), offer predictive advantage over widely used classical kernels such as the Radial Basis Function (RBF) (Kakavand et al., 2026).

However, recent literature has revealed a significant gap between this promise and the available empirical evidence. Bowles et al. (2024) demonstrated that many QML benchmarks lack strong baselines and adequate control of simplistic claims. Kakavand et al. (2026) conducted the most comprehensive study to date—with 970 experiments, nested cross-validation, and IBM hardware validation—and concluded that quantum kernels for tabular data rarely outperform well-tuned classical baselines. Heyraud et al. (2022) characterised the geometric distortion induced by noise in quantum kernel machines, showing that the effective rank and spectrum of the kernel matrix are substantially altered by physical noise channels. Yin et al. (2025) demonstrated superiority on a photonic processor, but on a platform and task that differs from the standard tabular flow. Sahin et al. (2025) proposed approximation techniques (KTA and Nyström) to reduce evaluation cost, but without integrating paired corrected inference into the decision flow. Active Quantum Kernel Acquisition (2026) optimised anchor-pair acquisition and shot budgets, but presupposes that progression to QPU has already been authorised.

In this landscape, a clear methodological gap emerges: no prospective and sequential rule currently exists that integrates criteria of statistical effect, equivalence, geometric survival through fidelity tiers, computational cost, and hardware validation into a transparent go/no-go decision structure. Existing studies treat these criteria in a fragmented manner—either focusing on benchmarks without corrected inference, or spectral diagnostics without fidelity progression, or cost optimisation without a decision gate.

The present article contributes the **Geometry-Survival Evidence Gate**—a pre-specified decision rule that answers the following operational question: *should one proceed from simulation to quantum hardware on a given classification task?* The Evidence Gate integrates five sequential criteria: (i) statistical superiority of the quantum kernel over the classical baseline, assessed by Nadeau–Bengio corrected testing; (ii) practical equivalence within a pre-specified margin; (iii) survival of accuracy across three fidelity tiers—exact statevector, 2048-shot estimation, and Aer

depolarised noise simulation; (iv) complete computational cost accounting; and (v) anchor-pair validation with minimum-target correlation between ideal fidelity and QPU. The rule is binary (go/no-go) and was frozen via SHA-256 hash before experiment execution.

This study has two declared objectives. The first is to present and formalise the Evidence Gate as a reusable methodological contribution for the QML community. The second is to honestly report the negative result obtained: under the pre-specified protocol, the SVM with fidelity quantum kernel demonstrated statistical inferiority to the SVM-RBF on tabular data, with $\Delta\text{BAC} = -0.217$ (95% corrected CI $[-0.285; -0.148]$). This result is restrictive—it does not settle the question of quantum kernel utility in general, but constrains future hypotheses and demonstrates the applicability of the Evidence Gate as a quality control mechanism for QML research.

Section 2 presents the theoretical background on quantum kernels, nested cross-validation, and corrected statistical inference. Section 3 describes the experimental protocol. Section 4 reports the results. Section 5 discusses implications, limitations, and future work. Section 6 concludes.

2 Theoretical Background

2.1 Quantum kernels and feature maps

A quantum kernel is defined by the similarity function between two data points, computed via the inner overlap of quantum states prepared by a parametrised circuit (Kakavand et al., 2026):

$$K(\mathbf{x}, \mathbf{y}) = |\langle \phi(\mathbf{x}) | \phi(\mathbf{y}) \rangle|^2, \quad (1)$$

where $|\phi(\mathbf{x})\rangle$ is the quantum state prepared by the feature map from the classical data vector $\mathbf{x}$. The ZZFeatureMap used in this study applies R_z rotations with angular data encoding followed by controlled-Z entangling gates between pairs of qubits, producing interference that captures pairwise feature correlations (Havlíček et al., 2019). For two qubits with one repetition and linear entanglement, the circuit is:

$$U_\phi(\mathbf{x}) = \exp\left(i \sum_S \phi_S(\mathbf{x}) \prod_{k \in S} Z_k\right) H^{\otimes n}, \quad (2)$$

where S indexes subsets of qubits with entanglement and ϕ_S are data encoding functions. The kernel matrix K is evaluated by computing $|\langle \phi(\mathbf{x}_i) | \phi(\mathbf{x}_j) \rangle|^2$ for all pairs (i, j) in the dataset, producing a positive semi-definite matrix that feeds the SVM (Havlíček et al., 2019; Kakavand et al., 2026).

Exact evaluation of the kernel matrix requires inner overlap computation via statevector (perfect quantum simulation). On real hardware, the estimate is obtained by sampling (shots): each data pair is prepared as a quantum state, measured repeatedly, and fidelity is estimated by outcome counting (Heyraud et al., 2022). The variance of this estimate scales as $O(1/N_{\text{shots}})$, introducing statistical noise that adds to hardware physical noise.

2.2 Noise and geometric degradation

On real quantum hardware, three error sources affect the kernel matrix: (i) gate errors, which distort the prepared state; (ii) readout errors, which alter measured outcomes; and (iii) decoherence, which destroys quantum correlations (Heyraud et al., 2022). Modelable noise channels—such as the depolarising channel with parameter p —transform the ideal state ρ_0 into:

$$\mathcal{E}(\rho) = (1 - p) \rho_0 + \frac{p}{2^n} I, \quad (3)$$

where n is the number of qubits and I is the identity matrix. This distortion alters the kernel matrix spectrum, reducing kernel-target alignment and modifying the effective rank (Heyraud et al., 2022; Sahin et al., 2025).

The concept of **geometric survival**—the maintenance of predictive accuracy across successive fidelity tiers (statevector $\rightarrow$ shots $\rightarrow$ noise)—is central to this study. Survival is not guaranteed: shot statistical fluctuations can alter classification decisions, and physical noise can degrade the kernel structure (Heyraud et al., 2022). The validity ladder quantifies this survival empirically.

2.3 Nested cross-validation and corrected inference

Cross-validation (CV) is the standard for generalisation estimation in machine learning (Refsgaard et al., 1999). However, standard CV violates the independence assumption between folds, inflating statistical power and producing underestimated confidence intervals (Nadeau and Bengio, 2003). Nested CV partially addresses this problem by separating hyperparameter selection (inner folds) from generalisation estimation (outer folds) (Refsgaard et al., 1999).

Nadeau and Bengio (2003) proposed a correction for the standard error in dependent CV estimates:

$$SE_{\text{corr}} = SE \times \sqrt{\frac{1}{n_{\text{test}}} + \frac{n_{\text{test}}}{n_{\text{train}}}}, \quad (4)$$

where n_{test} and n_{train} are the test and training set sizes, respectively. This correction is conservative: it produces wider confidence intervals that adequately reflect the dependence between folds.

The primary one-sided test— H_1 : paired mean $\Delta\text{BAC} > 0$ —is evaluated with the Nadeau–Bengio corrected t -statistic. Sensitivity analyses include: (i) exact sign permutation, testing H_0 : $\text{median}(\Delta) = 0$ without normality assumption; (ii) TOST (Two One-Sided Tests) equivalence with pre-specified margin ± 0.02 in BAC; and (iii) Holm adjustment for secondary outcomes.

2.4 Evidence gates in reproducible science

The concept of an evidence gate has roots in the clinical trials literature and open science practices, where go/no-go criteria are pre-specified before data collection (Scheel et al., 2022). The Evidence Gate proposed in this study adapts this concept to QML, integrating statistical, geometric, and economic criteria into a formal decision sequence.

The distinction between the Evidence Gate and existing approaches is important. Kakavand et al. (2026) conducted rigorous benchmarks, but without a formal decision rule for when to advance to hardware. Active Quantum Kernel Acquisition (2026) optimised resource allocation, but presupposes prior advancement. Bowles et al. (2024) cautioned against simplistic claims, but did not provide an operational decision structure. The Evidence Gate fills this gap by formalising the question: *based on simulated evidence, should one reserve QPU time for this task?*

3 Methods

3.1 Experimental design

The study was designed as a comparative computational investigation, pre-specified and frozen via SHA-256 hash before execution. The primary outcome was the paired difference in balanced accuracy ($\Delta\text{BAC} = \text{BAC}_{\text{QML}} - \text{BAC}_{\text{RBF}}$). The pre-specified classification of the result was: superiority, inferiority, practical equivalence, or inconclusive.

Four tabular datasets were used: (i) `make_moons` (synthetic, 200 samples, 2 features, Gaussian noise $\sigma = 0.1$); (ii) Iris binary (100 samples, 4 features, classes 0/1); (iii) Wine binary (130 samples, 13 features, classes 0/1); (iv) Breast Cancer Wisconsin (180 samples, 30 features, classes malignant/benign). All datasets were binarised and subjected to identical preprocessing: missing-value imputation (median), standardisation (z -score), and dimensionality reduction by PCA to 2 components. All preprocessing was fit exclusively on the training set in each fold to prevent data leakage.

3.2 Repeated nested cross-validation

The validation design used repeated nested CV with: (i) 4 stratified outer folds (generalisation estimation); (ii) 3 outer repetitions with different seeds; (iii) 3 stratified inner folds (hyperparameter selection). The total was 12 outer evaluations per dataset.

In each outer fold, the imputation–standardisation–PCA–angular scaling transformation was fit exclusively on the training data. Internal selection used a hyperparameter grid: $C \in \{0.1, 1.0, 10.0\}$ for both SVMs and $\gamma \in \{\text{scale}, \text{auto}\}$ for the RBF. The winning internal classifier was selected by mean balanced accuracy on inner folds and then re-evaluated on the outer fold.

3.3 Quantum kernel

The quantum feature map used ZZFeatureMap with 2 qubits, 1 repetition, and linear entanglement. The kernel matrix was evaluated across three fidelity tiers (validity ladder):

1. **Exact statevector:** direct inner overlap computation without sampling—represents the ideal noise-free limit.
2. **2048 shots:** sampling-based estimation with 2048 independent measurements per pair—introduces statistical fluctuation.
3. **Aer with noise:** simulation of the Qiskit Aer depolarising channel with parameters $p_1 = 0.001$ (single-qubit gate error), $p_2 = 0.01$ (two-qubit gate error) and $p_{\text{readout}} = 0.02$ (readout error)—represents typical NISQ hardware conditions.

3.4 Evidence Gate—decision rule

The Evidence Gate is a sequential decision rule with five pre-specified criteria. The protocol advances through stages only if **all** criteria of the current stage are satisfied:

Stage 1 — Primary inference:

- ΔBAC corrected mean > 0 (statistical superiority)
- 95% bilateral corrected CI (Nadeau–Bengio) above zero
- Classification: inferiority, equivalence, or inconclusive $\rightarrow$ **stop**

Stage 2 — Practical equivalence:

- TOST with margin ± 0.02 in BAC yields $p < 0.05$
- If not equivalent $\rightarrow$ continue only if Stage 3 is promising

Stage 3 — Geometric survival:

- BAC at exact statevector $\geq$ classical baseline
- BAC with 2048 shots $\geq$ 95% of statevector BAC
- BAC with Aer noise $\geq$ 90% of statevector BAC
- Drop $> 10\%$ across any transition $\rightarrow$ **stop**

Stage 4 — Computational cost:

- Kernel time recorded and comparable to baseline
- Estimated logical evaluations reported

- Infeasible cost $\rightarrow$ **stop**

Stage 5 — Anchor pairs and QPU advancement:

- Pearson/Spearman correlation ideal-QPU ≥ 0.90 on anchor pairs
- MAE ≤ 0.10 between ideal fidelity and QPU on anchor pairs
- Budgetary cost viability
- All satisfied $\rightarrow$ **go** for full QPU classifier

3.5 Statistical inference

The primary test used Nadeau–Bengio correction for fold dependence (Nadeau and Bengio, 2003), with $\alpha = 0.05$ and one-sided test ($H_1: \mu_{\Delta\text{BAC}} > 0$). Pre-specified sensitivity analyses: (i) exact sign permutation with 1024 permutations; (ii) TOST equivalence with margin ± 0.02 ; (iii) paired Cohen’s d_z ; (iv) 95% bootstrap interval (5000 resamples); (v) Holm adjustment for secondary outcomes.

3.6 Kernel diagnostics

Three diagnostic metrics were recorded at each evaluation: (i) kernel–target alignment (Bowles et al., 2024); (ii) effective rank, defined as $\exp(-\sum_i p_i \ln p_i)$ where $p_i = \lambda_i / \sum_j \lambda_j$ (Heyraud et al., 2022); (iii) relative Frobenius error between the evaluated kernel matrix and the statevector reference.

3.7 Claim control

Claim control rules were pre-specified (Bowles et al., 2024; Kakavand et al., 2026):

- Do not claim “first study” without an updated systematic review.
- Do not claim “quantum advantage” from simulator accuracy.
- Distinguish predictive superiority, practical equivalence, geometric survival, and computational advantage.
- Report null and negative results without softening.
- Identify as exploratory any analysis not frozen in the SHA-256 protocol.

4 Results

4.1 Primary result—statistical inferiority

In repeated nested cross-validation (12 outer evaluations, `make_moons` dataset), the mean primary outcome was $\Delta\text{BAC} = -0.217$, with Nadeau–Bengio corrected 95% CI $[-0.285; -0.148]$ and corrected one-sided $p = 0.9999$ (Table 1). The confidence interval is entirely negative, indicating that the SVM-RBF outperforms the fidelity quantum kernel in balanced accuracy within this protocol.

The paired Cohen’s $d_z = -4.495$, indicating a large-magnitude difference between models. This value reflects the consistency of the pattern: across all 12 outer evaluations, the SVM-RBF outperformed the quantum kernel.

Table 1: Statistical inference results for the primary and secondary outcomes.

Outcome	Mean	95% CI Lower	95% CI Upper	Corrected p
Δ balanced accuracy (primary)	-0.217	-0.285	-0.148	0.9999
Δ accuracy	-0.217	-0.285	-0.148	2.38×10^{-5}
$\Delta F1$	-0.200	-0.296	-0.104	8.05×10^{-4}

Table 2: Pre-specified sensitivity analyses.

Analysis	Statistic	p / CI	Interpretation
Nadeau–Bengio (one-sided H_1 : $\Delta > 0$)	Corrected t	$p = 0.9999$	No superiority evidence
Sign permutation (bilateral)	Exact p	$p = 0.0005$	Significant inferiority
TOST equivalence (± 0.02)	One-sided p	$p = 0.9999$	Not equivalent
Cohen’s d_z	$d_z = -4.495$	—	Large negative effect
95% bootstrap Δ BAC (5000)	Median	$[-0.563; 0.063]$	Includes zero (uncorrected)
Fraction positive Δ resamples	—	4.2%	Rarely does quantum kernel win

4.2 Sensitivity analyses

Exact sign permutation (1024 permutations) yielded $p = 0.0005$, confirming statistical inferiority. TOST equivalence with margin ± 0.02 yielded $p = 0.9999$, indicating the models are *not* equivalent.

4.3 Robust validation—outer repetitions

Table 3 presents the 12 outer evaluations.

The SVM-RBF won in 12/12 evaluations (100%), with mean BAC of 0.955 (SD 0.050). The quantum kernel showed mean BAC of 0.738 (SD 0.047). The mean Δ BAC was -0.217 (SD 0.048).

4.4 Validity ladder—geometric survival

Geometric survival was **complete**: BAC remained stable at 0.625 across all tiers. Relative Frobenius error grew progressively ($0.000 \rightarrow 0.011 \rightarrow 0.055$), indicating increasing kernel distortion, but this did not alter SVM classification decisions.

4.5 Computational cost

Kernel time ranged from 0.89 s to 1.44 s (mean 1.01 s). The 2048-shot evaluation required 23.27 s ($423 \times$ statevector), and Aer noise required 8.59 s ($156 \times$ statevector). Estimated logical evaluations: $1,008 \times 2048 = 2,064,384$ shots.

4.6 Model comparison

4.7 Multi-dataset suite—exploratory analysis

Across all 16 folds, Δ BAC was negative. Geometric survival was high (> 0.93), but did not translate into predictive advantage.

Table 3: Repeated nested cross-validation results (12 outer evaluations).

Run Kernel (s)	Rep.	Fold	Seed	BAC RBF	BAC QML	Δ BAC	C_{RBF}	C_{QML}
1 0.89	1	1	1051	1.000	0.767	−0.233	10.0	1.0
2 0.91	1	2	2060	0.933	0.733	−0.200	10.0	0.1
3 0.92	1	3	3069	0.967	0.767	−0.200	10.0	0.1
4 0.91	1	4	4078	0.933	0.800	−0.133	10.0	0.1
5 0.96	2	1	5087	0.967	0.767	−0.200	10.0	1.0
6 0.95	2	2	6096	0.967	0.667	−0.300	10.0	10.0
7 1.12	2	3	7105	0.900	0.667	−0.233	10.0	0.1
8 1.29	2	4	8114	0.967	0.733	−0.233	1.0	1.0
9 1.44	3	1	9123	1.000	0.700	−0.300	10.0	10.0
10 0.94	3	2	10132	0.833	0.667	−0.167	1.0	10.0
11 0.89	3	3	11141	1.000	0.800	−0.200	10.0	10.0
12 0.95	3	4	12150	0.933	0.733	−0.200	10.0	1.0

Table 4: Validity ladder—statevector $\rightarrow$ shots $\rightarrow$ Aer noise.

Level	Accuracy	$F1$	Frobenius	Alignment	Eff. Rank	Time (s)	Errors
0 · Exact SV	0.625	0.500	0.000	0.109	8.63	0.055	—
1 · 2048 shots	0.625	0.500	0.011	0.110	8.93	23.271	—
2 · Aer noise	0.625	0.500	0.055	0.110	10.34	8.595	0.001/0.01/0.02

Table 5: Model comparison—make_moons (single split).

Model	Accuracy	BAC	$F1$	Kernel (s)	95% Bootstrap CI
SVM-RBF	0.875	0.875	0.889	—	[0.688; 1.000]
Logistic Regression	0.813	0.813	0.842	—	[0.625; 1.000]
SVM + Quantum Kernel	0.625	0.625	0.500	23.27	[0.375; 0.875]

Table 6: Multi-dataset application suite (exploratory, not frozen).

Dataset	Fold	ΔBAC	Geom. Surv.	Frobenius	Alignment	Rel. Rank	Kernel (s)
make_moons	1	-0.300	0.978	0.022	0.093	0.117	0.86
make_moons	2	-0.200	0.980	0.020	0.105	0.115	0.89
make_moons	3	-0.133	0.980	0.020	0.122	0.112	0.89
make_moons	4	-0.167	0.976	0.024	0.093	0.111	0.90
iris_binaria	1	-0.119	0.979	0.021	0.130	0.115	0.68
iris_binaria	2	-0.237	0.936	0.064	0.147	0.127	0.63
iris_binaria	3	-0.080	0.981	0.019	0.134	0.116	0.64
iris_binaria	4	-0.237	0.983	0.017	0.166	0.116	0.66
wine_binario	1	-0.072	0.978	0.022	0.167	0.099	1.03
wine_binario	2	-0.028	0.978	0.022	0.261	0.095	1.40
wine_binario	3	-0.059	0.984	0.016	0.224	0.096	1.77
wine_binario	4	-0.127	0.975	0.025	0.128	0.109	1.05
breast_cancer	1	-0.133	0.984	0.016	0.189	0.074	1.55
breast_cancer	2	-0.020	0.985	0.015	0.200	0.073	1.53
breast_cancer	3	-0.087	0.985	0.015	0.199	0.074	1.52
breast_cancer	4	-0.053	0.984	0.016	0.195	0.078	1.49

Table 7: Exploratory Spearman correlations between kernel metrics and ΔBAC .

Metric	ρ (Spearman)	p	Interpretation
Geometric survival $\rightarrow \Delta\text{BAC}$	0.415	0.110	Moderate, non-significant
Kernel-target alignment $\rightarrow \Delta\text{BAC}$	0.739	0.001	Strong, significant

Table 8: Quantum kernel diagnostics.

Metric	Value
Maximum asymmetry (eigenvalues)	2.22×10^{-16}
Maximum diagonal deviation	0.006
Smallest eigenvalue	-5.97×10^{-16}
Kernel-target alignment	0.110
Diagonal tolerance	0.022

Table 9: Evidence Gate sequential evaluation.

Criterion	Status	Result
Stage 1: $\Delta\text{BAC} > 0$, $\text{CI}_{95\%} > 0$	FAIL	$\Delta\text{BAC} = -0.217$; CI $[-0.285; -0.148]$
Stage 2: Equivalence (TOST, ± 0.02)	FAIL	$p = 0.9999$ (not equivalent)
Stage 3: Geometric survival	PASS	BAC stable 0.625 across 3 tiers
Stage 4: Computational cost	PASS	Kernel ~ 1 s; cost recorded
Stage 5: Anchor-pair QPU	NOT ASSESSED	Conditional on Stage 1

4.8 Mechanistic analysis

4.9 Kernel diagnostics

4.10 Evidence Gate result

Evidence Gate classification: NO-GO—inconclusive for QPU advancement.

5 Discussion

5.1 Statistical inferiority

The primary outcome—statistical inferiority of the fidelity quantum kernel over the SVM-RBF ($\Delta\text{BAC} = -0.217$)—is consistent with recent literature (Bowles et al., 2024; Kakavand et al., 2026). The RBF baseline was tuned by nested validation, representing a strong classical competitor. The quantum kernel showed consistently low kernel–target alignment (0.093–0.261). This geometric inadequacy—not noise instability—appears to be the dominant factor in inferiority.

The spectral diagnostics (Table 8) reveal that the 2-qubit feature map may be structurally insufficient to capture class separability in tabular data with 2–30 features. The low relative effective rank (7–12%) indicates underutilisation of circuit capacity.

5.2 Geometric survival without predictive advantage

A conceptual contribution is the empirical demonstration that **geometric survival** $\neq$ **predictive advantage**. The kernel survived perfectly across statevector $\rightarrow$ shots $\rightarrow$ Aer, but its performance was consistently inferior to the RBF. Survival indicates noise robustness; inferiority indicates geometric inadequacy.

5.3 Correlation between alignment and performance

The strong correlation between kernel–target alignment and ΔBAC ($\rho = 0.739$, $p = 0.001$) is consistent with kernel theory. However, even the highest observed alignment (0.261) was insufficient to produce positive ΔBAC , suggesting an alignment threshold below which the quantum kernel offers no advantage.

5.4 Positioning relative to the literature

The present study distinguishes itself in three aspects: (i) the Evidence Gate as a methodological contribution; (ii) transparent negative result (Scheel et al., 2022); (iii) cost integration as a decision factor.

5.5 Limitations

- (i) Small scale: 200 samples, 90/10 splits.
- (ii) Limited feature map: 2 qubits, 1 repetition.
- (iii) Simulation only (Aer), not real QPU.
- (iv) Tabular data only; may differ for molecular/materials data.
- (v) PCA to 2 components may discard discriminative information.
- (vi) Residual fold dependence despite correction.
- (vii) Mechanistic analyses are exploratory (not frozen in SHA-256).

5.6 Future work

- (i) Feature map expansion to identify alignment threshold for advancement.
- (ii) QPU validation with anchor pairs (Criteria 3–5).
- (iii) Cost reduction via approximation (Nyström, KTA).
- (iv) Multi-class and regression extensions.
- (v) Quantum-structure data (molecular, materials).
- (vi) Systematic review with retrospective Evidence Gate application.

5.7 Implications for QML practice

- (i) A strong baseline is essential; weak baselines produce superficial superiority claims.
- (ii) Kernel diagnostics (alignment, effective rank, Frobenius error) must be routinely reported.
- (iii) Negative results are valuable and constrain future hypotheses.
- (iv) Evidence Gates should be adopted as minimum standard for QML evaluation.

6 Conclusions

This study presented the **Geometry-Survival Evidence Gate**—a prospective, sequential go/no-go decision rule for quantum kernel application on tabular data—and honestly reported the negative result obtained under a pre-specified protocol.

The SVM with fidelity quantum kernel demonstrated **statistical inferiority** to the SVM-RBF in repeated nested cross-validation ($\Delta\text{BAC} = -0.217$; 95% corrected CI $[-0.285; -0.148]$; one-sided $p = 0.9999$). The kernel survived geometrically across three fidelity tiers (statevector $\rightarrow$ shots $\rightarrow$ Aer noise), but this survival did not translate into predictive advantage. The correlation between kernel–target alignment and ΔBAC ($\rho = 0.739$) suggests that feature map geometry is the limiting factor, not noise robustness.

The Evidence Gate classified this protocol as **NO-GO** for QPU advancement, with failure at Criteria 1 (superiority) and 2 (equivalence). The result is restrictive: it does not settle quantum kernel utility in general, but constrains hypotheses for tabular data with low-capacity feature maps and demonstrates the Evidence Gate as a quality control mechanism for QML research.

It is recommended that the community adopt prospective evidence gates as minimum standard for quantum kernel evaluation, valuing transparent negative results and promoting reproducibility in an area prone to premature claims.

References

- Active Quantum Kernel Acquisition (2026). Active quantum kernel acquisition under a shot budget. *arXiv preprint*.
- Bowles, J., Ahmed, S., and Schuld, M. (2024). Better than classical? the subtle art of benchmarking quantum machine learning models. *arXiv preprint*.
- Havlíček, V. et al. (2019). Supervised learning with quantum-enhanced feature spaces. *Nature*, 567:209–212.
- Heyraud, V. et al. (2022). Noisy quantum kernel machines. *Physical Review A*, 106:052421.

- Kakavand, S., Strohmeier, C., and Schlotter, M. (2026). Benchmarking quantum kernel support vector machines against classical baselines on tabular data. *arXiv preprint*.
- Nadeau, C. and Bengio, Y. (2003). Inference for the generalization error. *Machine Learning*, 52(3):239–281.
- Refsgaard, A. et al. (1999). On the proper use of cross-validation for prediction and model selection in hydrology. *Hydrology and Earth System Sciences*, 3(3):403–413.
- Sahin, E. et al. (2025). Quantum-efficient kernel target alignment. *arXiv preprint*.
- Scheel, A. M., Tiokin, L., Schönfeld, T., and Christopoulos, V. (2022). The role of negative results in science. *MetaArXiv*.
- Yin, Z. et al. (2025). Experimental quantum-enhanced kernel-based machine learning on a photonic processor. *Nature Photonics*, 19:1020–1027.

A Experiment Configuration

Parameter	Value
Researcher	Marcelo Claro Laranjeira
Seed	42
Shots	2048
Feature map	ZZFeatureMap
Feature map reps	1
Entanglement	Linear
C (quantum SVM)	1.0
Kernel time (full split)	23.27 s
Kernel–target alignment	0.110
Diagonal tolerance	0.022
Protocol SHA-256	89c08f07...
Python	3.12.13
NumPy	2.0.2
Pandas	2.2.2
Scikit-learn	1.6.1
Qiskit	2.3.1
Qiskit Aer	0.17.2
Qiskit ML	0.9.0

B Pre-registered Protocol

The full protocol was frozen via SHA-256 and is available in `data/processed/protocolo_sha256.txt`. The mandatory execution order was: OSF $\rightarrow$ Nested CV $\rightarrow$ Application suite $\rightarrow$ Aer noise $\rightarrow$ Anchor pairs QPU $\rightarrow$ Full QPU classifier. The present execution completed items 1–3 and stopped at Evidence Gate Criterion 1 (statistical inferiority).